使用SnapVault 进行备份复制

NetApp Solutions SAP

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目录

使用SnapVault 进行备份复制 1
概述—使用SnapVault 备份复制 1
在ONTAP 文件系统的FSX上配置复制关系 2
将备份SVM添加到SnapCenter 7
为备份复制创建新的SnapCenter 策略 8
将策略添加到资源保护中 10
使用复制创建备份 11
从二级存储还原和恢复 15

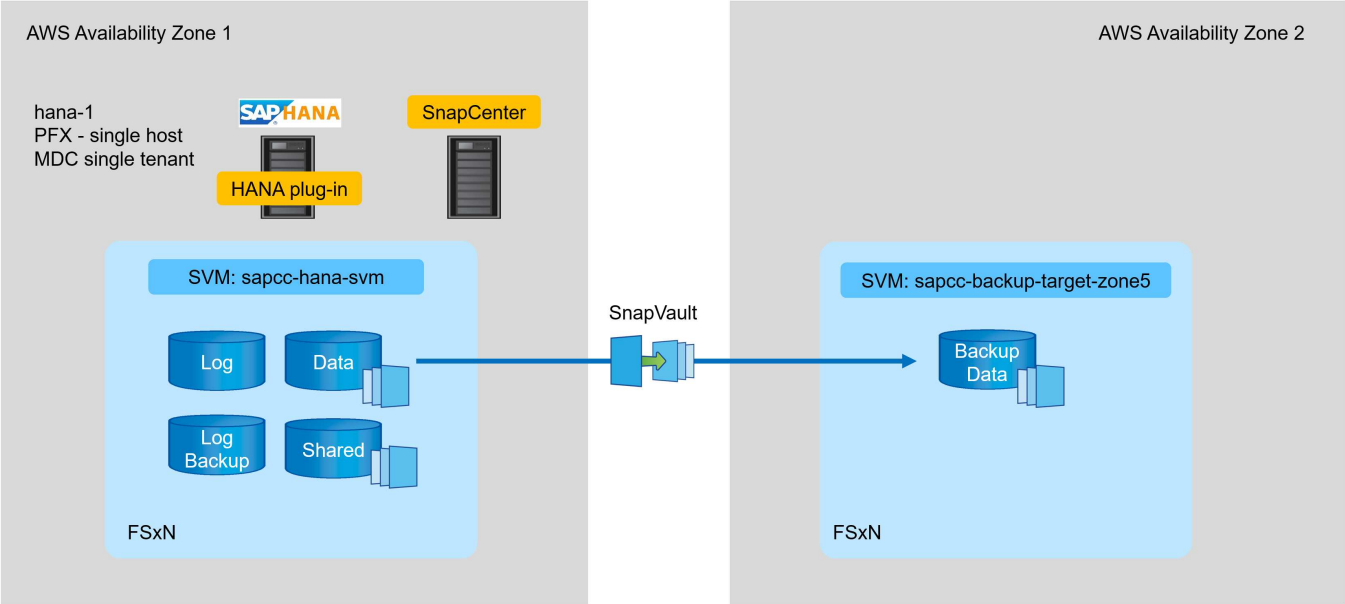
使用SnapVault 进行备份复制

概述—使用SnapVault 备份复制

"Previous：备份非数据卷。"

在我们的实验室设置中、我们会在第二个AWS可用性区域中使用第二个FSX for ONTAP 文件系统来展示HANA数据卷的备份复制。

如第章所述 ""数据保护策略""、复制目标必须是另一个可用性区域中ONTAP 文件系统的第二个FSx、才能防止ONTAP 文件系统的主FSx发生故障。此外、应将HANA共享卷复制到二级FSX for ONTAP 文件系统。



配置步骤概述

必须在适用于ONTAP 的FSX层上执行几个配置步骤。您可以使用NetApp Cloud Manager或FSX for ONTAP 命令行执行此操作。

1. 适用于ONTAP 文件系统的对等FSX。必须为适用于ONTAP 文件系统的FSX建立对等关系、才能在彼此之间进行复制。
2. 对等SVM。SVM必须建立对等关系、才能在彼此之间进行复制。
3. 创建目标卷。在目标SVM上创建卷、卷类型为`DP`。要用作复制目标卷、必须键入`DP`。
4. 创建SnapMirror策略。此选项用于创建类型为`vault`的复制策略。
   a. 向策略添加规则。此规则包含SnapMirror标签以及二级站点备份的保留。您必须稍后在SnapCenter 策略中配置相同的SnapMirror标签、以便SnapCenter 在包含此标签的源卷上创建Snapshot备份。
5. 创建SnapMirror关系。定义源卷和目标卷之间的复制关系并附加策略。
6. 初始化SnapMirror。此时将开始初始复制、在此复制中、完整的源数据将传输到目标卷。

卷复制配置完成后、您必须按如下所示在SnapCenter 中配置备份复制：

1. 将目标SVM添加到SnapCenter。
2. 为Snapshot备份和SnapVault 复制创建新的SnapCenter 策略。
3. 将此策略添加到HANA资源保护中。
4. 现在、您可以使用新策略执行备份。

以下各章将更详细地介绍各个步骤。

"下一步：在ONTAP 文件系统的FSX上配置复制关系。"

在ONTAP 文件系统的FSX上配置复制关系

"上一步：概述—使用SnapVault 备份复制。"

有关SnapMirror配置选项的追加信息 、请参见ONTAP 文档、网址为 "SnapMirror复制工作流(netapp.com)"。

- ONTAP 文件系统的源FSx：`FSxId00fa9e3c784b6abbb`
- 源SVM：`sAPCC—HANA—SVM`
- ONTAP 文件系统的目标FSX：`FSxId05f7f00af49dc7a3e`
- 目标SVM：`sAPCP-backup-target-zone5`

适用于ONTAP 文件系统的对等FSX

```
FsxId00fa9e3c784b6abbb::> network interface show -role intercluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
FsxId00fa9e3c784b6abbb
            inter_1      up/up    10.1.1.57/24
FsxId00fa9e3c784b6abbb-01
                                                                   e0e
true
            inter_2      up/up    10.1.2.7/24
FsxId00fa9e3c784b6abbb-02
                                                                   e0e
true
2 entries were displayed.
```

```
FsxId05f7f00af49dc7a3e::> network interface show -role intercluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
FsxId05f7f00af49dc7a3e
            inter_1      up/up    10.1.2.144/24
FsxId05f7f00af49dc7a3e-01
                                                                   e0e
true
            inter_2      up/up    10.1.2.69/24
FsxId05f7f00af49dc7a3e-02
                                                                   e0e
true
2 entries were displayed.
```

```
FsxId05f7f00af49dc7a3e::> cluster peer create -address-family ipv4 -peer
-addrs 10.1.1.57, 10.1.2.7
Notice: Use a generated passphrase or choose a passphrase of 8 or more
characters. To ensure the authenticity of the peering relationship, use a
phrase or sequence of characters that would be hard to guess.
Enter the passphrase:
Confirm the passphrase:
Notice: Now use the same passphrase in the "cluster peer create" command
in the other cluster.
```

`peer-addrs`是目标集群的集群IP。

```
FsxId00fa9e3c784b6abbb::>  cluster peer create -address-family ipv4 -peer
-addrs  10.1.2.144, 10.1.2.69
Notice: Use a generated passphrase or choose a passphrase of 8 or more
characters. To ensure the authenticity of the peering relationship, use a
        phrase or sequence of characters that would be hard to guess.
Enter the passphrase:
Confirm the passphrase:
FsxId00fa9e3c784b6abbb::>
FsxId00fa9e3c784b6abbb::> cluster peer show
Peer Cluster Name         Cluster Serial Number Availability
Authentication
------------------------- --------------------- --------------
--------------
FsxId05f7f00af49dc7a3e    1-80-000011           Available      ok
```

对等SVM

```
FsxId05f7f00af49dc7a3e::> vserver peer create -vserver sapcc-backup-
target-zone5 -peer-vserver sapcc-hana-svm -peer-cluster
FsxId00fa9e3c784b6abbb -applications snapmirror
Info: [Job 41] 'vserver peer create' job queued
```

```
FsxId00fa9e3c784b6abbb::> vserver peer accept -vserver sapcc-hana-svm
-peer-vserver sapcc-backup-target-zone5
Info: [Job 960] 'vserver peer accept' job queued
```

```
FsxId05f7f00af49dc7a3e::> vserver peer show
            Peer        Peer                            Peering
Remote
Vserver     Vserver     State        Peer Cluster      Applications
Vserver
----------- ----------- ------------ ----------------- --------------
---------
sapcc-backup-target-zone5
            peer-source-cluster
                        peered       FsxId00fa9e3c784b6abbb
                                                       snapmirror
sapcc-hana-svm
```

创建目标卷

您必须创建类型为`DP`的目标卷、以将其标记为复制目标。

```
FsxId05f7f00af49dc7a3e::> volume create -vserver sapcc-backup-target-zone5
-volume PFX_data_mnt00001 -aggregate aggr1 -size 100GB -state online
-policy default -type DP -autosize-mode grow_shrink -snapshot-policy none
-foreground true -tiering-policy all -anti-ransomware-state disabled
[Job 42] Job succeeded: Successful
```

创建 **SnapMirror** 策略

SnapMirror策略和添加的规则用于定义保留和SnapMirror标签、以确定应复制的快照。稍后创建SnapCenter 策略时、必须使用相同的标签。

```
FsxId05f7f00af49dc7a3e::> snapmirror policy create -policy snapcenter-
policy -tries 8 -transfer-priority normal -ignore-atime false -restart
always -type vault -vserver sapcc-backup-target-zone5
```

```
FsxId05f7f00af49dc7a3e::> snapmirror policy add-rule -vserver sapcc-
backup-target-zone5  -policy snapcenter-policy -snapmirror-label
snapcenter -keep 14
```

```
FsxId00fa9e3c784b6abbb::> snapmirror policy showVserver Policy
Policy Number         Transfer
Name    Name               Type   Of Rules Tries Priority Comment
------- ------------------ ------ -------- ----- -------- ----------
FsxId00fa9e3c784b6abbb
        snapcenter-policy  vault         1     8  normal  -
  SnapMirror Label: snapcenter                        Keep:      14
                                                Total Keep:      14
```

创建**SnapMirror**关系

现在、源卷和目标卷之间的关系已定义、XDP类型以及我们先前创建的策略也已定义。

```
FsxId05f7f00af49dc7a3e::> snapmirror create -source-path sapcc-hana-
svm:PFX_data_mnt00001 -destination-path sapcc-backup-target-
zone5:PFX_data_mnt00001 -vserver sapcc-backup-target-zone5 -throttle
unlimited -identity-preserve false -type XDP -policy snapcenter-policy
Operation succeeded: snapmirror create for the relationship with
destination "sapcc-backup-target-zone5:PFX_data_mnt00001".
```

初始化SnapMirror

使用此命令将开始初始复制。这是从源卷到目标卷的所有数据的完整传输。

```
FsxId05f7f00af49dc7a3e::> snapmirror initialize -destination-path sapcc-
backup-target-zone5:PFX_data_mnt00001 -source-path sapcc-hana-
svm:PFX_data_mnt00001
Operation is queued: snapmirror initialize of destination "sapcc-backup-
target-zone5:PFX_data_mnt00001".
```

您可以使用`snapmirror show`命令检查复制的状态。

```
FsxId05f7f00af49dc7a3e::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
sapcc-hana-svm:PFX_data_mnt00001
            XDP  sapcc-backup-target-zone5:PFX_data_mnt00001
                             Uninitialized
                                     Transferring   1009MB    true
02/24 12:34:28
```

```
FsxId05f7f00af49dc7a3e::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
sapcc-hana-svm:PFX_data_mnt00001
            XDP  sapcc-backup-target-zone5:PFX_data_mnt00001
                             Snapmirrored
                                     Idle           -         true    -
```

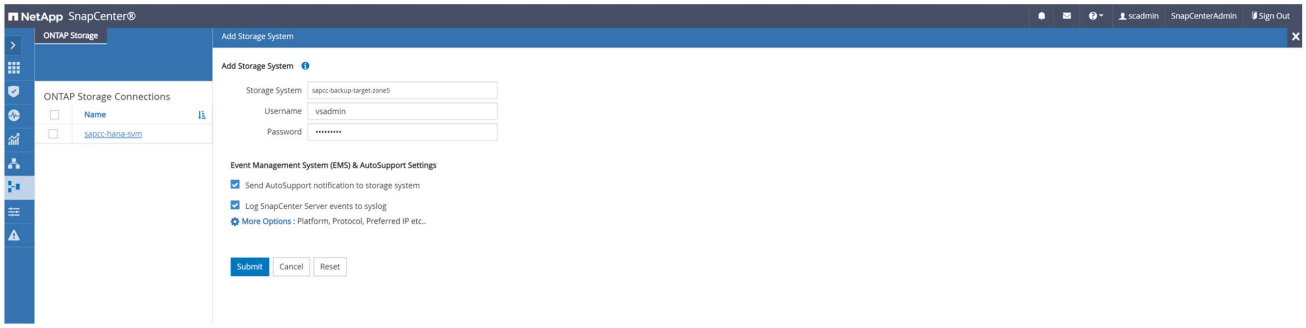
"下一步：将备份SVM添加到SnapCenter。"

将备份SVM添加到SnapCenter

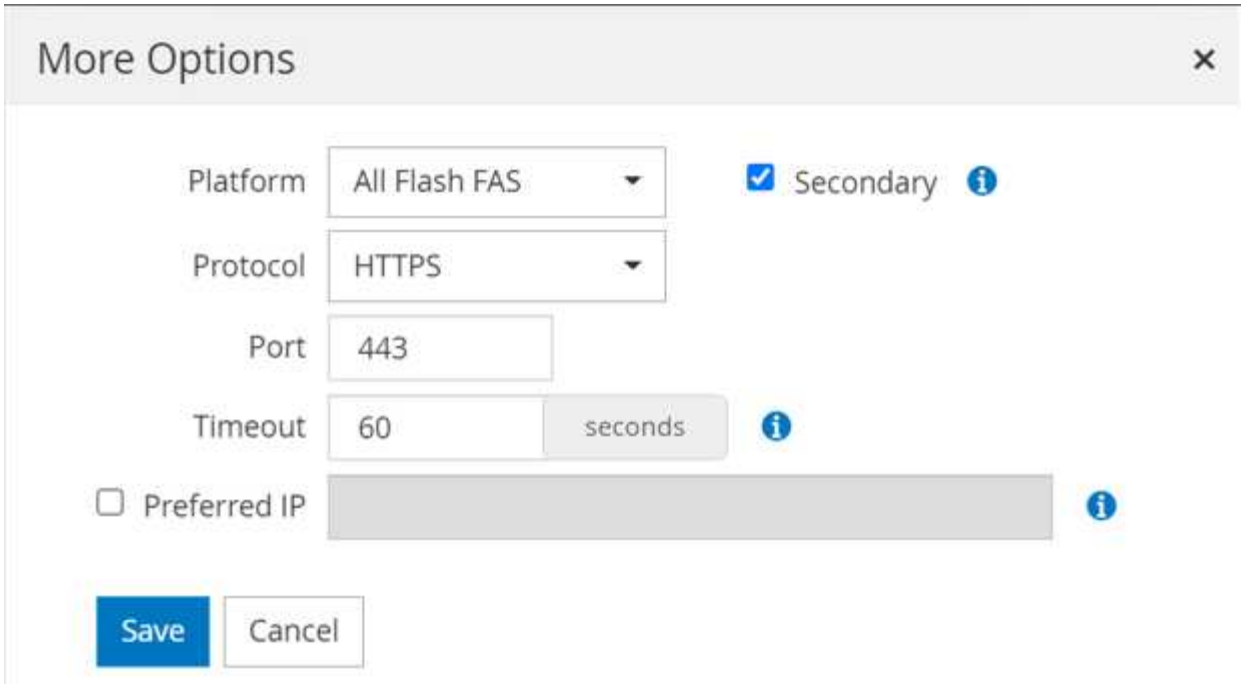
"先前版本：在ONTAP 文件系统的FSX上配置复制关系。"

要将备份SVM添加到SnapCenter 、请执行以下步骤：

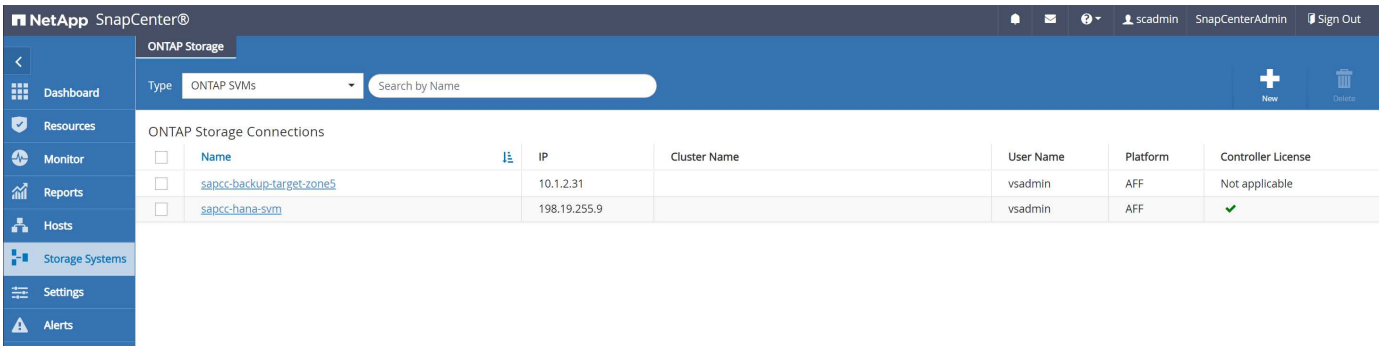
1. 在SnapCenter 中配置SnapVault 目标卷所在的SVM。



2. 在更多选项窗口中、选择全闪存FAS 作为平台、然后选择二级。



SVM现在可在SnapCenter 中使用。



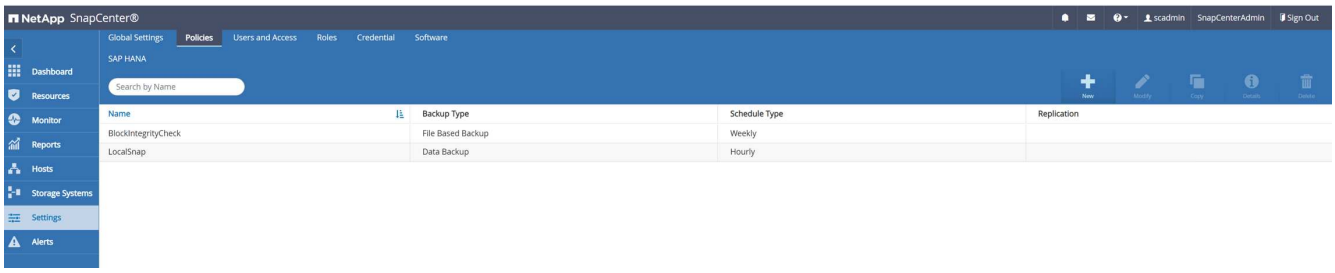
"下一步：创建新的SnapCenter 策略以进行备份复制。"

为备份复制创建新的SnapCenter 策略

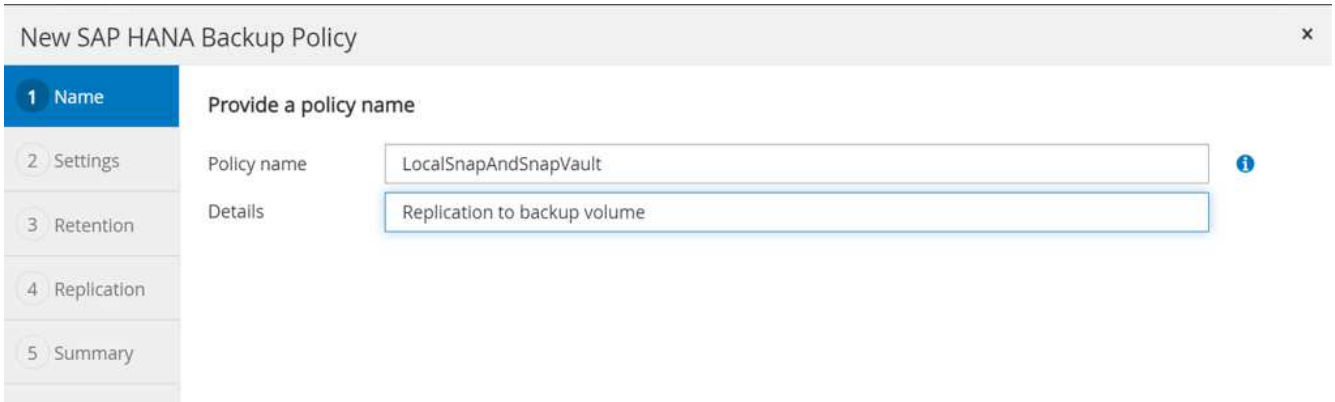
"上一步：将备份SVM添加到SnapCenter。"

您必须按如下所示配置备份复制策略：

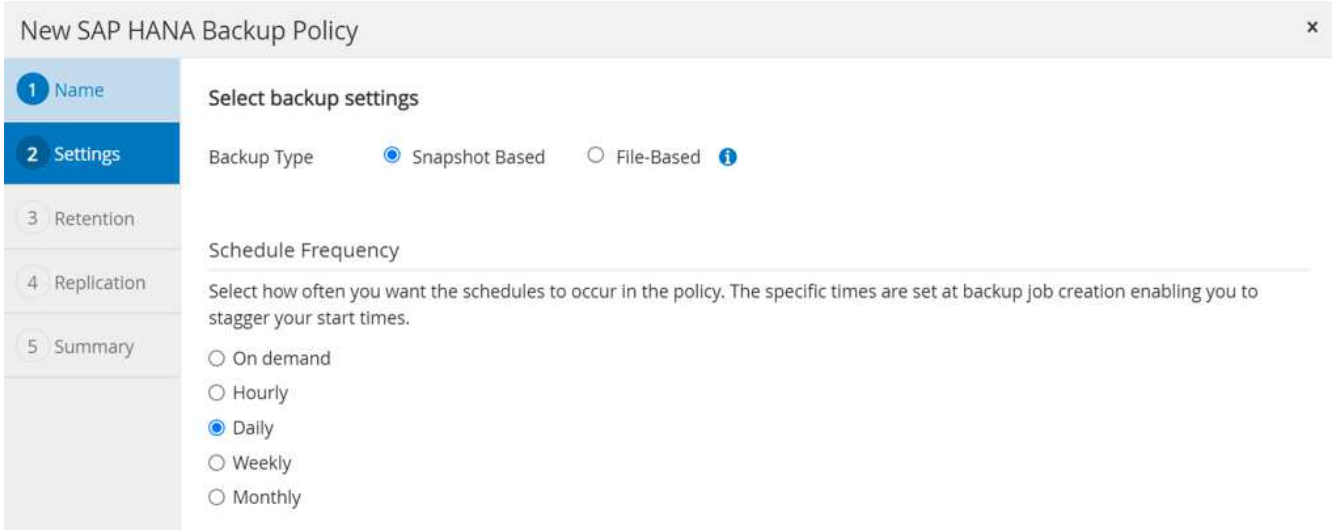
1. 请为此策略提供一个名称。



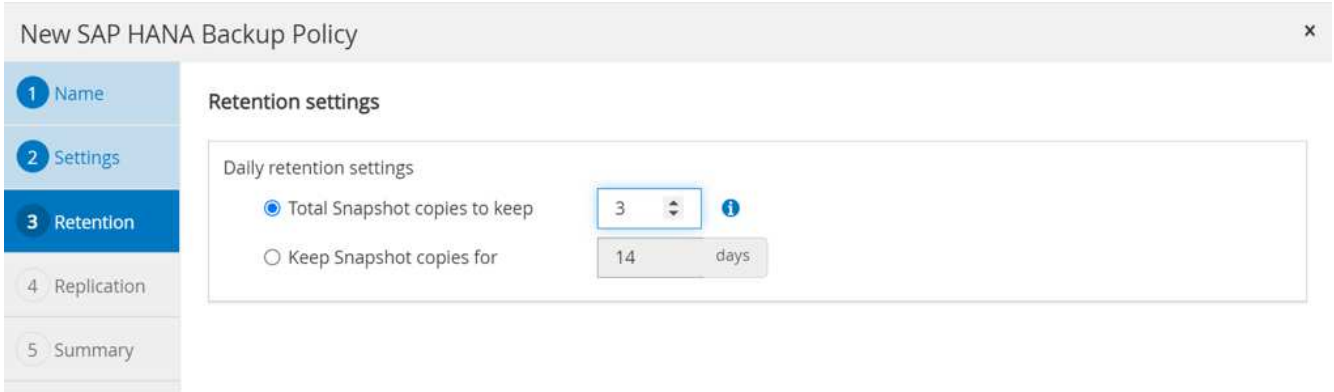
2. 选择Snapshot备份和计划频率。每天通常用于备份复制。



3. 选择Snapshot备份的保留。

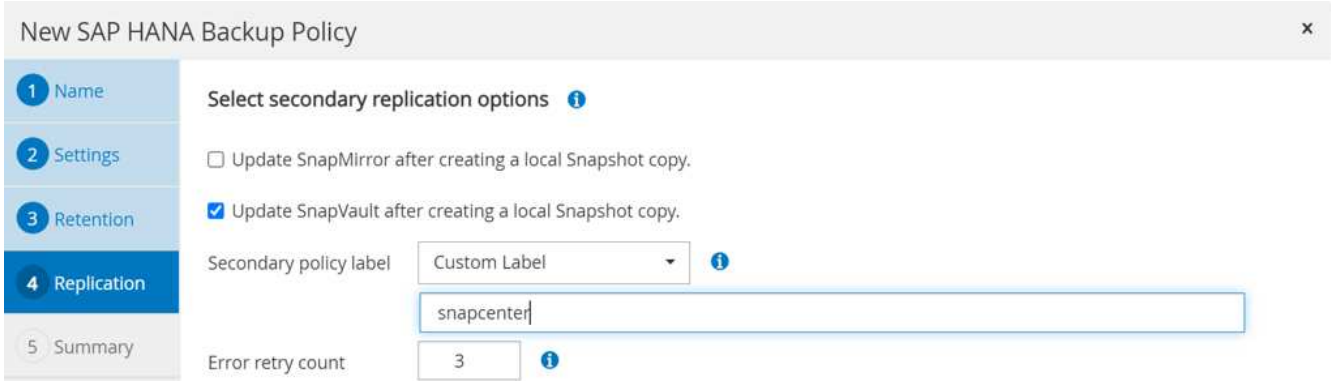


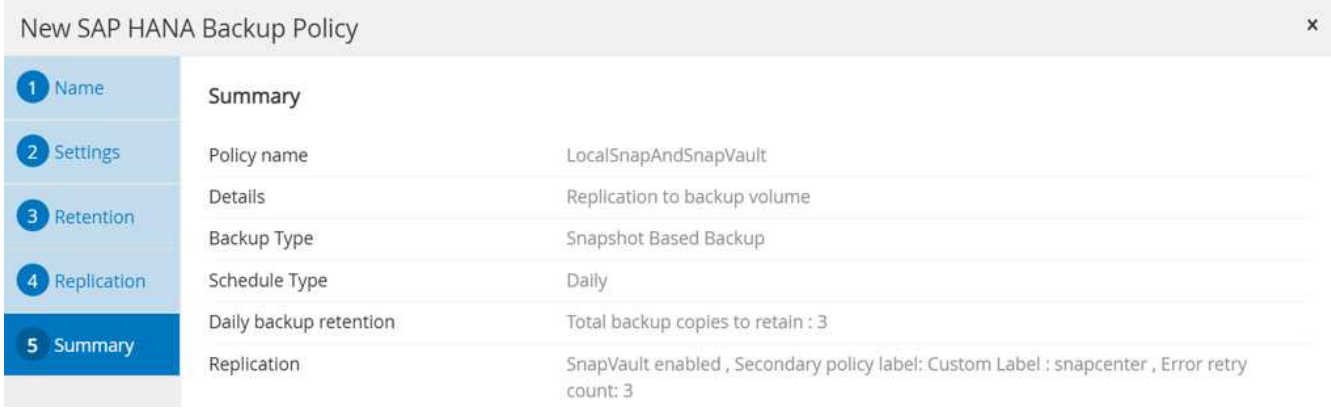
这是在主存储上创建的每日Snapshot备份的保留期限。先前已在SnapVault 目标上使用add rule命令在ONTAP 级别配置二级备份的保留。请参见《在适用于ONTAP 文件系统的FSX上配置复制关系》(xref)。



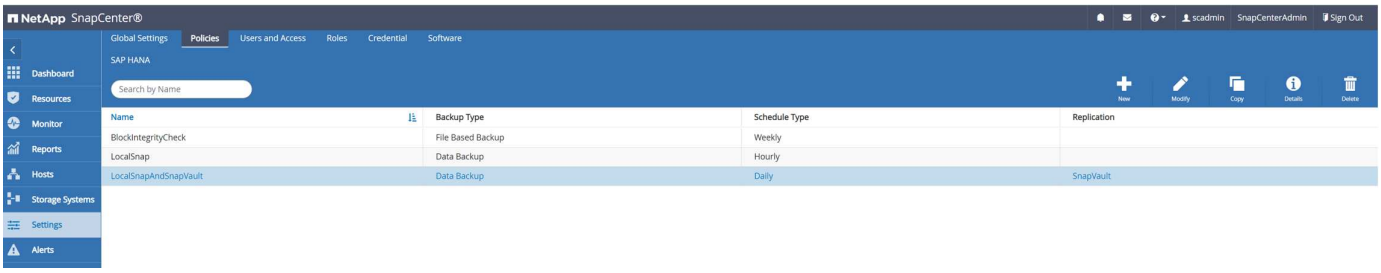
4. 选择Update SnapVault 字段并提供自定义标签。

   此标签必须与ONTAP 级别的`add rule`命令中提供的SnapMirror标签匹配。





此时将配置新的SnapCenter 策略。

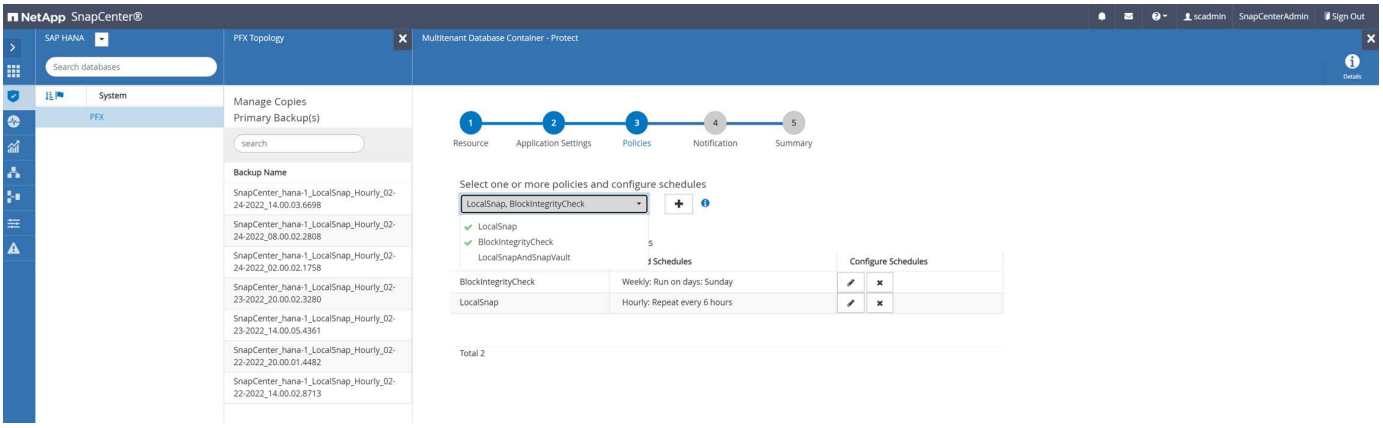


"下一步：将策略添加到资源保护中。"

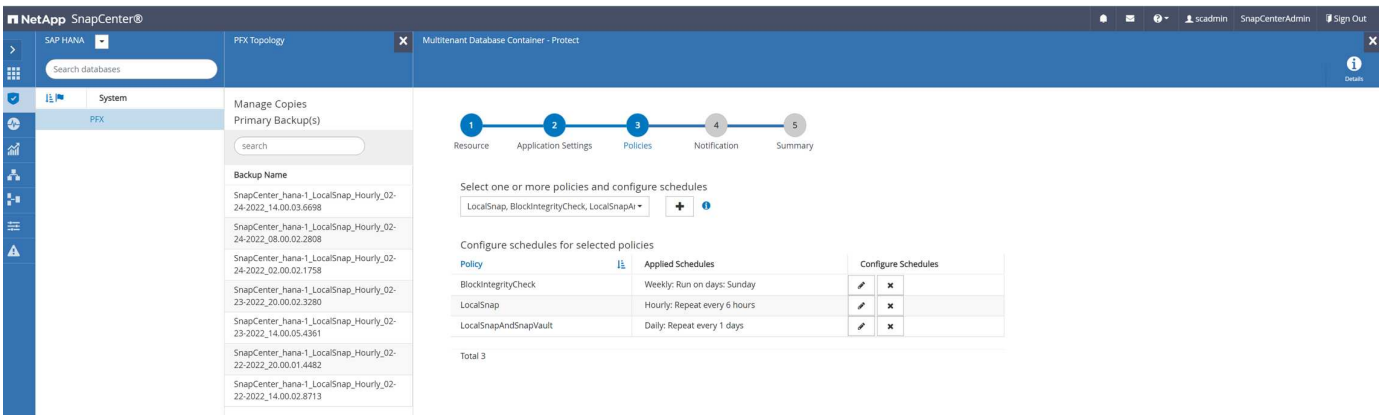
将策略添加到资源保护中

"先前：创建新的SnapCenter 策略以进行备份复制。"

您必须将新策略添加到HANA资源保护配置中、如下图所示。



我们在设置中定义了每日计划。



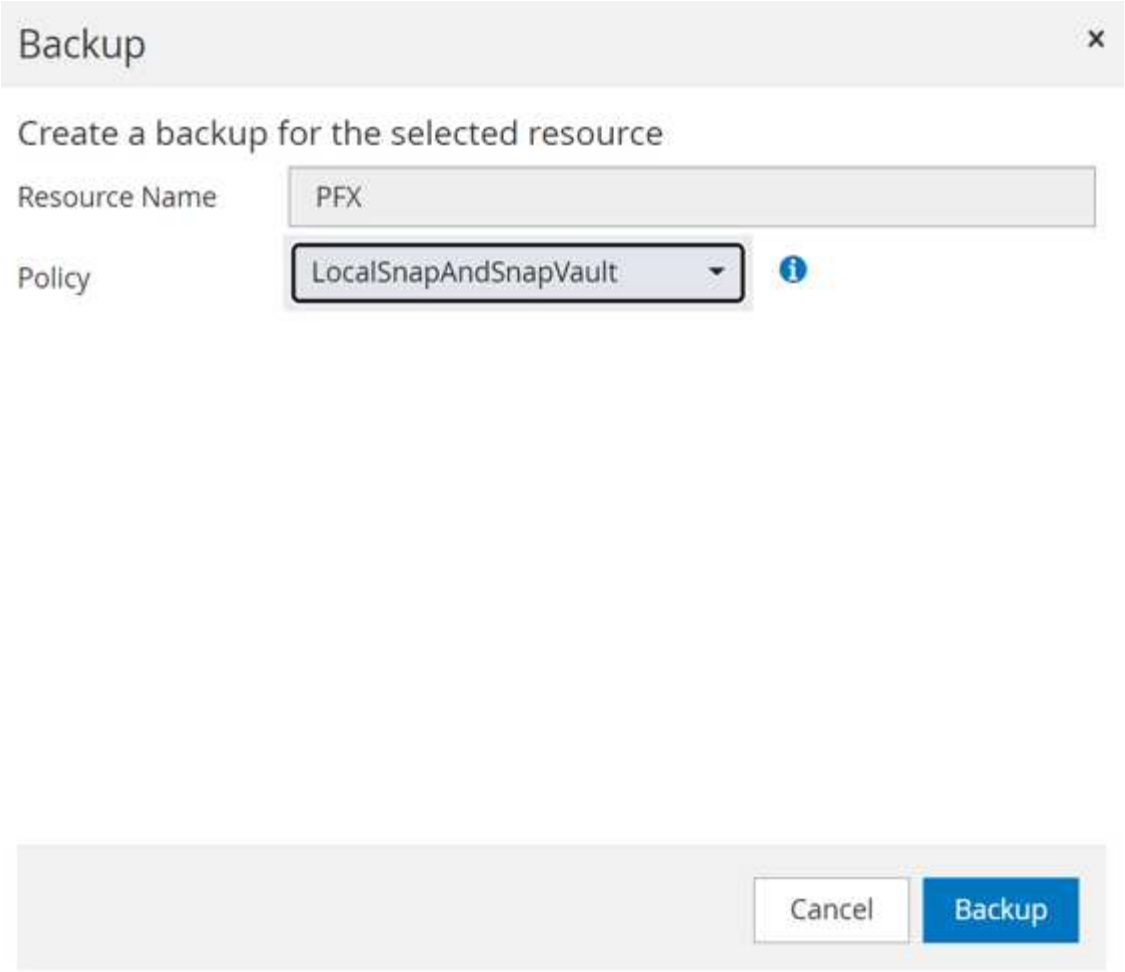
"下一步：使用复制创建备份。"

使用复制创建备份

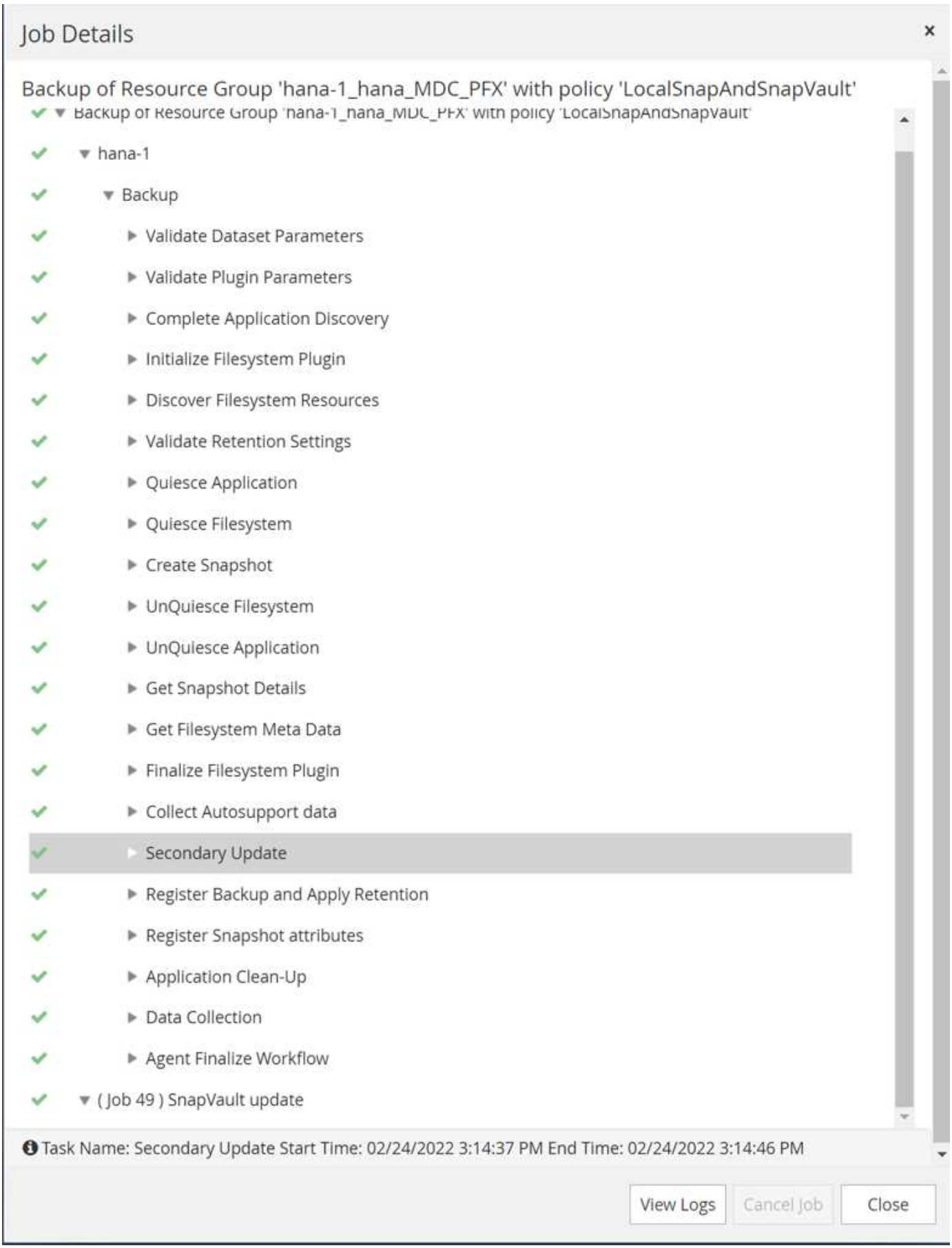
"上一步：将策略添加到资源保护中。"

创建备份的方式与创建本地Snapshot副本的方式相同。

要使用复制创建备份、请选择包含备份复制的策略、然后单击备份。



在SnapCenter 作业日志中、您可以看到二级更新步骤、该步骤将启动SnapVault 更新操作。将更改的块从源卷复制到目标卷。



在适用于ONTAP 的FSX文件系统上、使用SnapCenter 策略中配置的SnapMirror标签`SnapCenter`在源卷上创

建Snapshot。

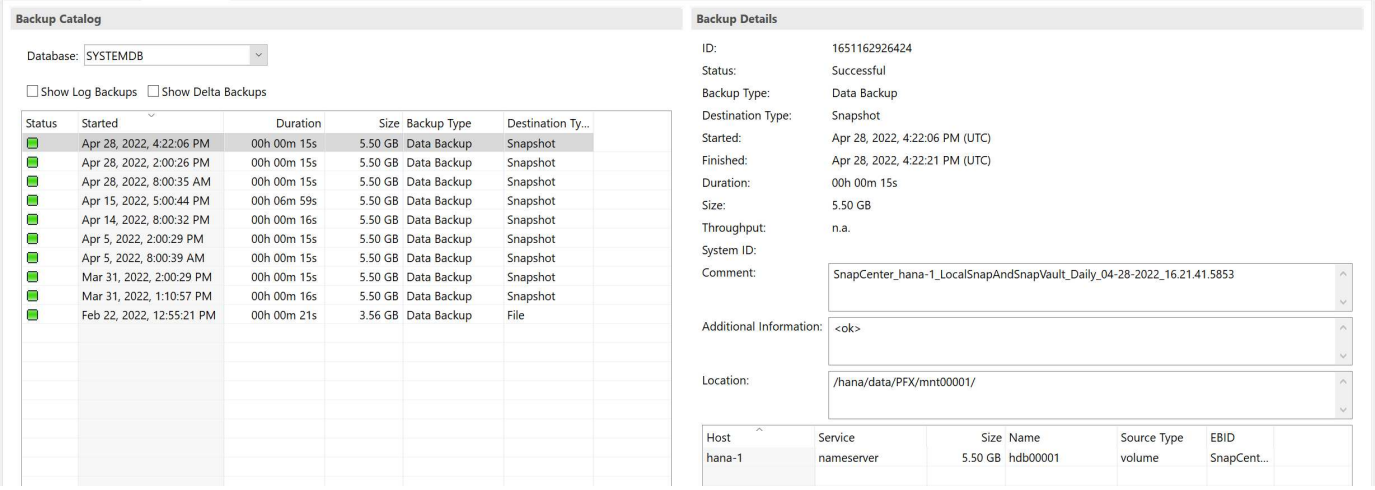
```
FsxId00fa9e3c784b6abbb::> snapshot show -vserver sapcc-hana-svm -volume
PFX_data_mnt00001 -fields snapmirror-label
vserver        volume            snapshot
snapmirror-label
-------------- -----------------
------------------------------------------------------------
----------------
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_03-31-
2022_13.10.26.5482 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_03-31-
2022_14.00.05.2023 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-05-
2022_08.00.06.3380 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-05-
2022_14.00.01.6482 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-14-
2022_20.00.05.0316 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-28-
2022_08.00.06.3629 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-1_LocalSnap_Hourly_04-28-
2022_14.00.01.7275 -
sapcc-hana-svm PFX_data_mnt00001 SnapCenter_hana-
1_LocalSnapAndSnapVault_Daily_04-28-2022_16.21.41.5853

snapcenter
8 entries were displayed.
```

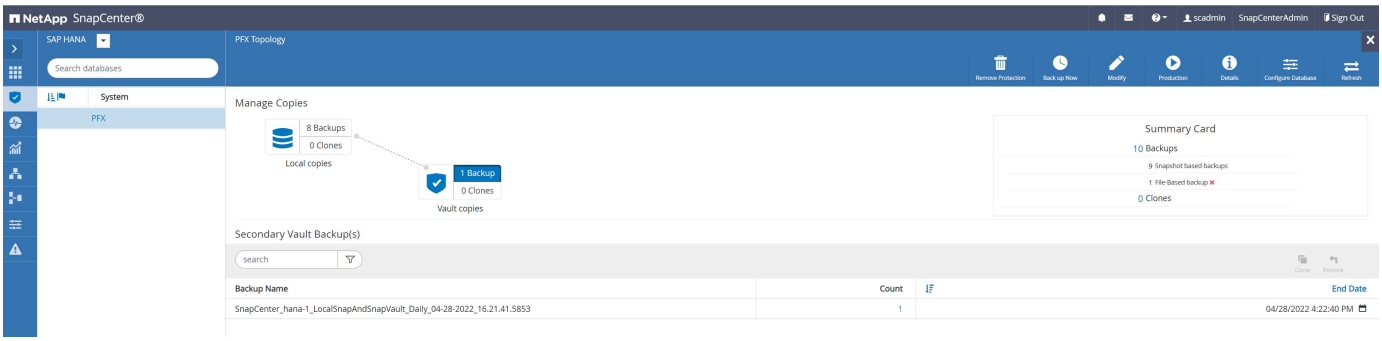
在目标卷上、将创建一个同名Snapshot副本。

```
FsxId05f7f00af49dc7a3e::> snapshot show -vserver sapcc-backup-target-zone5
-volume PFX_data_mnt00001 -fields snapmirror-label
vserver                   volume            snapshot
snapmirror-label
------------------------- -----------------
-------------------------------------------------------------------------
----------------
sapcc-backup-target-zone5 PFX_data_mnt00001 SnapCenter_hana-
1_LocalSnapAndSnapVault_Daily_04-28-2022_16.21.41.5853 snapcenter
FsxId05f7f00af49dc7a3e::>
```

新的Snapshot备份也会列在HANA备份目录中。



在SnapCenter 中、您可以通过单击拓扑视图中的存储副本来列出复制的备份。



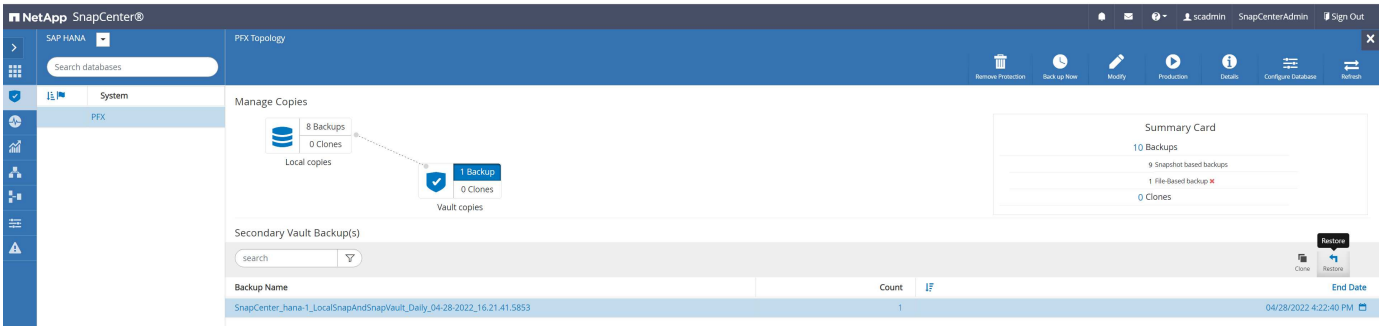
"下一步：从二级存储还原和恢复。"

从二级存储还原和恢复

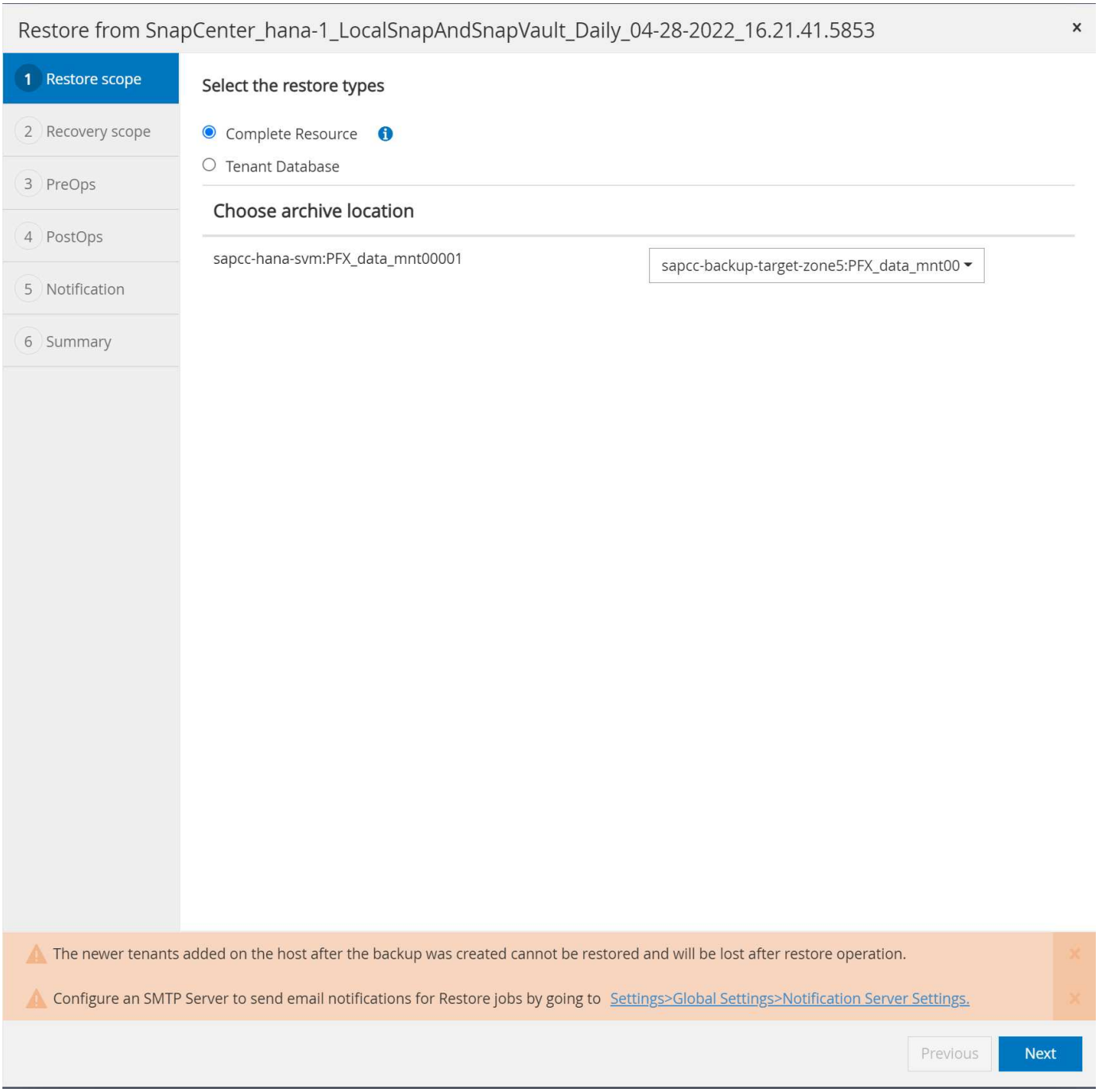
"上一步：使用复制创建备份。"

要从二级存储还原和恢复、请执行以下步骤：

要检索二级存储上所有备份的列表、请在SnapCenter 拓扑视图中单击存储副本、然后选择一个备份并单击还原。



还原对话框将显示二级位置。



进一步的还原和恢复步骤与先前在主存储上对Snapshot备份所述的步骤相同。

"下一步：从何处查找追加信息 和版本历史记录。"

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